Advaith Appajodu

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Education

University of California, Berkeley — Berkeley, CA
B.A. in Applied Mathematics — *Aug. 2025 – May 2029*

Relevant Coursework: CS61A (Current), Math 53 (Current)

Granite Bay High School — Granite Bay, CA
Class of 2025

Projects

Efficient Training of LLMs for Function Calling — San Francisco, CA
Salesforce AI Research — *Jan 2024 – Jun 2024*

- Researched LoRA fine-tuning methods to improve function discrimination in LLMs, achieving a measurable increase in function-call accuracy.
- Curated and preprocessed datasets from HuggingFace; conducted experiments tracking 50+ model runs via Weights & Biases.
- Authored a technical research paper summarizing methodology and findings under mentorship from Prof. Yingnian Wu (UCLA) and Erik Nijkamp (Salesforce AI).

Veritas AI Summer Cohort — Virtual
Veritas AI — *July 2023 – August 2023*

- Developed a convolutional neural network achieving >90% validation accuracy in diagnosing Malaria from image datasets.
- Created a predictive AI model estimating likelihood of car crashes using structured data (location, time, car type).
- Learned Python-based ML tools from mentors with Harvard and Brown affiliations; collaborated in a team of 6 on applied AI projects.

Experience

Computational Biology Program — Pittsburgh, PA
Carnegie Mellon University — *Jun 2024 – Jul 2024*

- Solved 10+ biological data challenges in daily hackathons (e.g., sequence alignment, DNA sequencing) improving computational biology skills.
- Performed wet lab work: extracted DNA, proteins, and bacteria from Acid Mine runoff samples.
- Explored metagenomics and species richness analysis; applied phylogenetic tree construction techniques.
- Led a research group of 4 in presenting findings on cancer treatment sequencing impacts to faculty and peers.

LCAP Committee Member — Roseville, CA
Roseville Joint Union High School District — *Sep 2023 – Present*

- Selected as 1 of 3 representatives by principal to present Granite Bay High School at the district level.
- Analyzed district-wide academic and financial data (>10,000 students impacted) to revise education plans for homeless and special needs students.
- Collaborated with administrators to design measurable benchmarks, improving tracking of district goals by 15%.

Technical Skills

Programming Languages: Python, Go, R, Java (basic)
ML/AI Frameworks: TensorFlow, PyTorch, Keras, HuggingFace, LoRA, WandB
Data Tools: Pandas, NumPy, dataset curation, preprocessing, visualization
Domains: Computational biology (DNA sequencing, metagenomics), function-calling LLMs
Other: Public speaking, project coordination, technical writing